

Plant Safety Protocol & Emergency Response Standard

1. Thermal and High-Voltage Hazards

CNC machines and automated presses operate with high-voltage 415V 3-phase power and hydraulic circuits exceeding 180 bar.

Extreme heat hazard: When a machine experiences thermal shutdown, spindle and cooling jacket temperatures can exceed 80 deg C. Personnel must wear thermal-resistant PPE and allow 45 minutes natural cooling before opening enclosure panels.

2. Lock-Out / Tag-Out (LOTO) Requirement

Before performing any mechanical inspection, filter replacement, or electrical servicing, technicians must apply padlocks and warning tags to the primary circuit breaker.

Never bypass interlock switches or coolant flow sensors during diagnosis.